

DEAKIN UNIVERSITY

RESEARCH PROJECT/RESEARCH PROJECT (ADVANCED)

ONTRACK SUBMISSION

Research Progress Check Point 1

Submitted By:

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Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO3 Communication of Research Outcomes

March 17, 2026



SIT724 / SIT746 Research Project – Task 3.1P

Research Progress Checkpoint 1

Student Name: Nikhil

Student ID: 224234862

Supervisor: Dr. Ananda Maiti

Project Title: Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

Unit: SIT724 Research Project / SIT746 Research Project (Advanced)

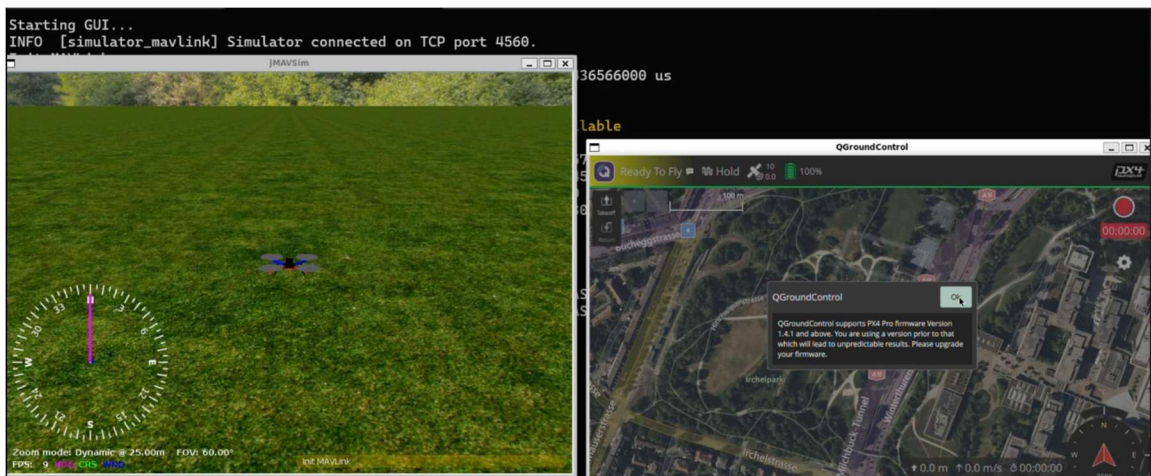
Trimester: Trimester 1, 2026

Overview of Research Continuity

This research project is a direct continuation of my previous trimester work completed under SIT723/SIT792. The prior trimester established a strong academic and technical foundation, including an extensive literature review, a validated research methodology, and the development of a fully operational PX4–Gazebo simulation environment. The current trimester focuses on extending this foundation into advanced GPS spoofing simulation and detection research.

Key Achievement: PX4–Gazebo Docker Simulation Environment

During the previous trimester, I successfully designed, implemented, and validated a fully containerised PX4–Gazebo simulation environment. This environment integrates PX4 SITL, jMAVSIM, MAVLink communication, and QGroundControl within a reproducible Docker-based architecture. The simulation environment remains fully functional and continues to serve as the core experimental platform for the current research.



Current Work: MAVLink-Based GPS Detection Module

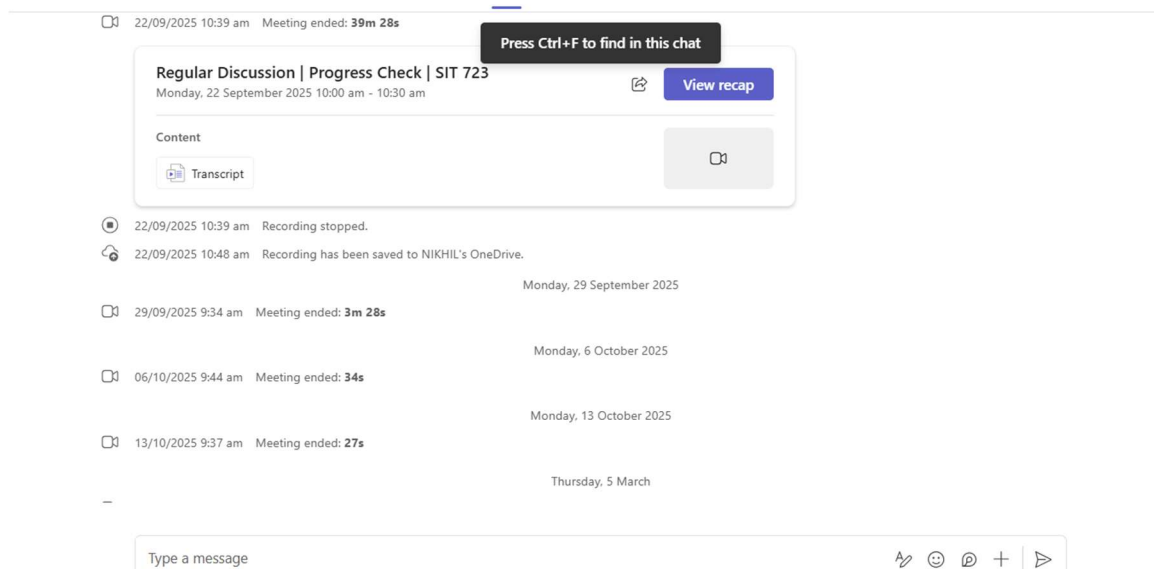
In the current trimester, I am actively developing a MAVLink-based GPS detection module within the existing simulation framework. This module is capable of real-time parsing of GPS telemetry data, extraction of latitude, longitude, altitude, and velocity information, and continuous logging for subsequent analysis. This functionality forms the sensing foundation for future spoofing detection algorithms.

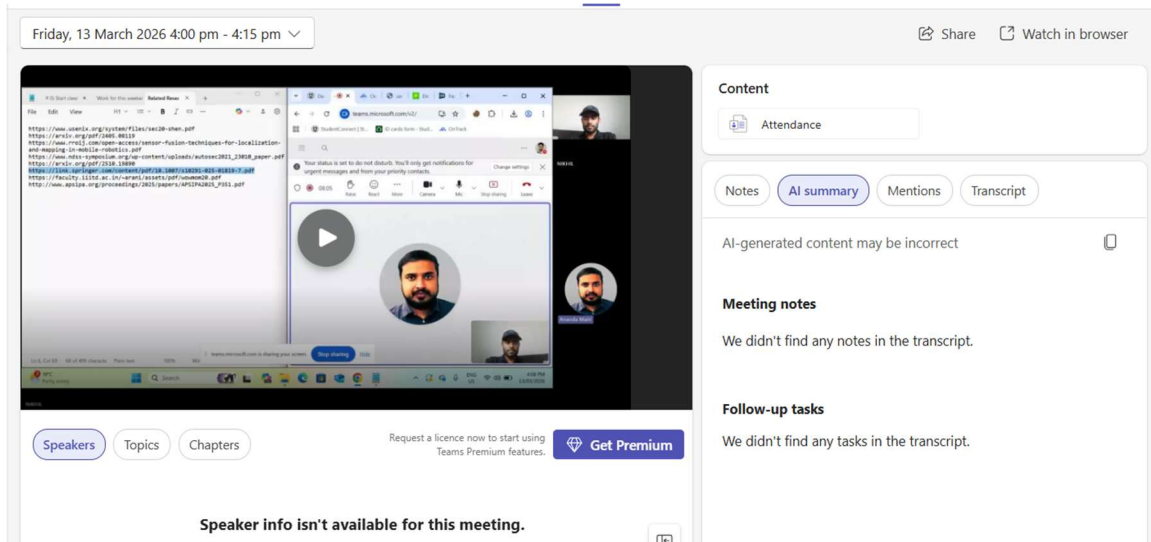
Research Focus: GPS Spoofing Simulation and Attack Models

Alongside implementation work, I am conducting focused research on GPS spoofing simulation and attack models. This includes analysis of sudden position jump attacks, gradual drift attacks, velocity manipulation, and trajectory inconsistencies. All attack modelling is strictly confined to simulated environments to ensure ethical and safe research practices.

Supervisor Meetings and Feedback

To date, I have conducted two supervision meetings with my mentor. These meetings focused on reviewing my simulation infrastructure, discussing progress on the MAVLink-based detection module, and refining the research direction for GPS spoofing attack simulation. Feedback from these meetings has been incorporated into my current implementation and research planning.





Current Challenges and Mitigation

The primary challenges at this stage include designing realistic spoofing scenarios without real signal transmission, managing false positives during dynamic vehicle motion, and ensuring scalability towards machine learning-based detection. These challenges are being addressed through controlled simulation-only attack modelling, sensor cross-validation, and incremental, modular system design.

Next Planned Activities

Upcoming work includes implementing controlled GPS spoofing injection within simulation, expanding the detection module to support advanced anomaly classification, **generating datasets for machine learning**, and continuing regular supervision meetings.

Conclusion

This progress checkpoint demonstrates that my research is progressing as planned. The project has successfully transitioned from infrastructure development in the previous trimester to active detection and attack modelling work. The current outcomes align with SIT724/SIT746 expectations and establish a strong basis for subsequent algorithm development.